The Intuition Laboratory: A Self-Improving Architecture for the Causal Interpretation of Situation Descriptions

Running Causal-Architect Against the Casebook of Causal Intuitions

Gerd Graßhoff iD
Humboldt-Universität zu Berlin, Forschungsinstitut des Deutschen Museums
gerd.grasshoff@hu-berlin.de

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Abstract

We report the first systematic run of the Causal-Architect inference engine against the philosophical casebook of causal intuitions, inside a learning environment — the Intuition Laboratory — that processes a natural-language description of an experimental or narrative setting, normalises it into factors, constructs the coincidence table, runs the regularity-theoretic analysis, judges the verdicts against the elicited intuitions, and, on a clash, enters a learning loop that improves the causal model through a fixed repertoire of epistemic operators (factor-set expansion, marking factors, typicality truncation). The intuition-lab benchmark comprises the twenty cases of the philosophical casebook's twenty-four Sprachspiele that lie within the implemented apparatus, carrying forty-nine layer-achievable intuition expectations. After two engine-level corrections that the loop itself forced — replacing Quine–McCluskey subcube minimisation by mosaic-relative minimisation, and minimum-cardinality by irredundant covers — the engine replicates 49 of 49 achievable expectations (100%): twelve cases by the core theory, three through learning operators, one partially at the contextualised layer, with four cases correctly identified as requiring theory layers not yet implemented. We describe the architecture, the two engine corrections and their theoretical significance, the evaluation protocol, and the limits of the reported success.

keywords regularity theory · minimal theories · causal intuitions · benchmark · learning architecture · coincidence analysis

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> **Note**
>
> **Causal Regularity Bundle — empirical validation.** This document is an auxiliary component of the *Causal Regularity Bundle*. It reports the systematic run of the `causal-architect` engine against the benchmark casebook (`vol3-casebook.qmd`) and the LK-99 paper audit, establishing the implementability of the standard stated in the foundational paper (`arxiv-causal-regularity.qmd`). Bundle concept and manifest: `bundle-hybrid-publication.qmd`, `bundle.yaml`.

1 Purpose

The casebook (vol3-casebook.qmd) fixed, for every benchmark case of the regularity-theoretic literature, a normalisation, a coincidence table, a derivation, and a verdict — all by hand, verified by scripts. The present paper closes the circle: the verdicts are now *computed* by the Causal-Architect engine itself, inside an environment that measures the system's capacity to interpret causal situations correctly and improves the causal model when it fails. The question posed is the operational form of the casebook's philosophical question: to what extent can a mosaic-based implementation of the minimal-theory framework (Graßhoff and May 2001; Baumgartner 2013a; Baumgartner and Falk 2023) recover the intuitive causal verdicts from tabular evidence alone — and what has to be added when it cannot?

2 Architecture

The Intuition Laboratory (`benchmark/intuition_lab/`) is organised as five components around the casebook's staged pipeline.

Normaliser (Stage I). Translates the natural-language situation into the factor set, the causal field, the token assignment, and the *process order* — the story's information about which factors are exogenous and which mediate. In the benchmark runs reported here the normalisations are the casebook's gold Stage-I panels; the NL-to-factors step itself is an agent role (a language model constrained by the normalisation rules of models.qmd), whose separate evaluation is future work (Section 6). The process order is genuine Stage-I output: stories state what acts first and through what; the engine receives it as data, not as a verdict.

Tabulator (Stage II). Projects the world — the case's full hidden coincidence table — onto the currently visible factor set, deduplicates, and monitors homogeneity. Contradictory configurations signal latent factors and hand control to the Learner before any verdict is attempted.

Engine (Stage III). `minimal_theory_search` of `causal_architect.mt`: mosaic-relative minimally sufficient conditions, irredundant covers as minimally necessary disjunctions, ambiguity sets, contradiction counts (both properties in their present form are results of this benchmark; Section 3). An adapter derives the two verdict types: *type-level relevance* as literal containment in some atomic minimal theory of the outcome — deliberately non-transitive, following (CR) (Baumgartner and Falk 2023) — and *actual causation* as containment plus an instantiated active route through direct theories (Baumgartner 2013a), where directness is resolved by the Stage-I process order (admissible covers lie

strictly upstream; the most proximate profile wins) and routes may not pass through biconditionals that mention factors already downstream on the path — the adapter's acyclicity stand-in for structural-redundancy elimination, which the engine does not yet implement.

Critic (Stage IV). The casebook's intuitions, encoded as expectations: candidate, outcome, expected verdict, verdict kind (type, token, unorientable), and the *theory layer* at which the expectation is achievable (MT, AC, cAC, AG, RT-B). The layer annotation is what makes the scoring honest: an engine implementing the Baumgartner layers must not be penalised for verdicts that the casebook itself shows to require the Andreas–Günther deviancy conditions or the Ramsey-test apparatus (Andreas and Günther 2024b, 2024a, 2019) — but it must be *flagged*, visibly, wherever that is the case.

Learner (Stage V). On inhomogeneity or clash, applies epistemic operators in a fixed order of increasing cost, re-running the pipeline after each application (at most six iterations):

1. **EXPAND** — reveal a latent factor offered by the environment; the operator implementing permanence (PERM): spurious relevance is unmasked by factor-set expansion.
2. **MARK** — introduce a marking factor with corrected observations, the normalisation repair that late preemption requires (models.qmd, Stage I).
3. **TYP** — typicality truncation of the coincidence table and re-minimisation: the contextualised layer cAC (Baumgartner 2013a, sec. 4), applied only where the case supplies a ranking.

A stable clash after all operators is classified as principled dissent (with the casebook's diagnosis) or, when the expectation is layer-flagged, as out of the implemented layers. Success categories: OK_CORE, OK_LEARNED, PARTIAL_LAYER, DISSENT, NOT_COVERED.

3 What the loop taught the engine

The most instructive results predate the final score: the first benchmark run failed on nine of sixteen covered cases, and the failures forced two corrections *of the engine* and two *of the verdict semantics* — each one a piece of regularity theory that the casebook states in prose and that the implementation had not honoured.

From Quine–McCluskey to mosaic-relative minimisation. The engine's minimisation merged subcubes à la Quine–McCluskey. QM presupposes that all 2^n configurations are empirically possible; on mosaics generated by causal structures this is false, and the engine consequently retained conjuncts that are redundant relative to the mosaic — its hooters theory read $M * T + \ldots \Rightarrow L$, keeping the hooters as a conjunct of five o'clock. This is precisely the defect Baumgartner (2013b) diagnosed in QCA's reliance on QM optimisation, reproduced in our own code. The fix implements minimal sufficiency as the casebook defines it: a conjunct is eliminable iff the reduced condition covers no *observed* negative row (`mosaic_msc_terms`).

From minimum-cardinality to irredundant covers. The engine returned only minimum-size covers. Minimal necessity, however, is *irredundancy* — no disjunct droppable — not minimum cardinality: in chain structures the direct theory $C + D \Rightarrow E$ is irredundant but larger than the root shortcut $A \Rightarrow E$ (wind-farm and switch structures), and discarding it costs exactly the verdicts that make the theory non-transitive and chain-aware (`all_irredundant_covers`). With both corrections in place, the

engine's own regression suite still passes, and the published fire example [Baumgartner 2009, Table 1] reproduces exactly: $SC * OX + FL * SP * OX \Rightarrow FI$.

Non-transitive relevance. The adapter's first relevance semantics closed relevance under chaining — and promptly declared the switch relevant to the arrival. Type-level relevance is containment in a minimal theory *of the outcome*, nothing more; the non-transitivity of the switch and short-circuit cases is a feature the semantics must preserve, not repair.

Direct theories and upstream biconditionals. Over small complete mosaics, root shortcuts, chain readings, and upstream biconditionals (the exogenous factor expressed through its own effects — the wind-farm structural-redundancy problem) are fit-equivalent. Token routes exist only through *direct* theories; the adapter resolves directness by the Stage-I process order and refuses routes through downstream-mentioning biconditionals. This is a stand-in: the principled solution is structural-redundancy elimination (SM) inside the engine (Baumgartner and Falk 2023), recorded as the engine's next milestone.

4 Evaluation protocol

Twenty cases; sixteen within the implemented layers (MT/CMT, AC, cAC); four outside them by the casebook's own analysis (extended double prevention and the assassin's mentor require the Andreas–Günther path and deviancy conditions; tower/shadow and forest fire the RT-B belief-revision apparatus). Within the sixteen, forty-nine expectations are achievable at the implemented layers and one is layer-flagged (the absence of poison in bogus prevention: no Baumgartner layer removes it; only AG deviancy does — the casebook's isomorphism analysis). Learning affordances exist in exactly three cases, each theoretically mandated: the epiphenomenon starts with the common cause unmeasured (EXPAND implements PERM); late preemption starts with the observationally naive table (MARK implements the marking factor); bogus prevention and Putin's omission carry typicality rankings (TYP implements cAC). Verdicts, operators, iterations, and logs are written to `benchmark/results/intuition-lab-results.json`; the run is reproducible with `PYTHONPATH=src python3 -m benchmark.intuition_lab.report`.

5 Results

Table 1 Intuition-lab benchmark over the casebook: 16 of 20 cases within the implemented layers; 12 replicated by the core engine, 3 after learning; achievable intuition replication 49/49 (rate 1.0).

Case	Layer	Category	Iter.	Operators	Intuitions (ok / layer-flagged / dissent)
Manchester Factory Hooters	MT	OK_CORE	1	—	3 / 0 / 0
Struck match	MT	OK_CORE	1	—	3 / 0 / 0

Case	Layer	Category	Iter.	Operators	Intuitions (ok / layer-flagged / dissent)
and sung song					
Smoking and lung cancer	MT	OK_CORE	1	—	3 / 0 / 0
Harold Bride's Havana	MT	OK_CORE	1	—	3 / 0 / 0
Epiphenomenon (latent common cause)	MT	OK_LEARNED	2	EXPAND(C)	3 / 0 / 0
Wind farm and nuclear plant	MT	OK_CORE	1	—	6 / 0 / 0
Causal chain	MT	OK_CORE	1	—	5 / 0 / 0
Symmetric overdetermination	AC	OK_CORE	1	—	2 / 0 / 0
Early preemption	AC	OK_CORE	1	—	3 / 0 / 0
Late preemption (marking factor)	AC	OK_LEARNED	2	MARK(BnE)	3 / 0 / 0
Switch (Flipper)	AC	OK_CORE	1	—	5 / 0 / 0
Short circuit (boulder),	AC	OK_CORE	1	—	3 / 0 / 0

Case	Layer	Category	Iter.	Operators	Intuitions (ok / layer-flagged / dissent)
neuron model					
Bogus prevention (typicality)	cAC	PARTIAL_LAYER	2	TYP	1 / 1 / 0
Omission — Putin (typicality)	cAC	OK_LEARNED	2	TYP	2 / 0 / 0
Omission — promising neighbour	AC	OK_CORE	1	—	2 / 0 / 0
Subcause (entangled causes)	AG	OK_CORE	1	—	2 / 0 / 0
Extended double prevention	AG	NOT_COVERED	0	—	—
Assassin's mentor	AG	NOT_COVERED	0	—	—
Tower and shadow ('because')	RT-B	NOT_COVERED	0	—	—
Forest fire ('because')	RT-B	NOT_COVERED	0	—	—

The success quality, stated precisely: **49 of 49 layer-achievable intuition expectations are replicated (100%)**. Twelve of sixteen covered cases succeed on the core theory without any operator — the complete type-level suite (hooters, match, contraposition, Havana, wind farm, chain) and, after the corrections of Section 3, overdetermination, early preemption, switch, boulder, the neighbour's omission, and the subcause. Three cases require exactly one learning operator each, and in every one the operator is the one the theory predicts. The epiphenomenon is solved by EXPAND: the initial view is inhomogeneous, and the revealed common cause eliminates the spurious sibling-effect cluster —

permanence enacted. Late preemption starts from the observationally naive table, wrongly certifies the preempted backup, and is repaired by MARK: the corrected observation distinguishes process stages, after which the route check blocks the backup — the marking factor enacted. Putin's omission is solved by TYP, reproducing the contextualised-cause analysis. Bogus prevention is PARTIAL_LAYER by design: TYP removes the antidote, as cAC predicts; the absence of poison remains a contextualised cause, exactly the residue the casebook attributes to the missing deviancy layer. There are no dissents: every clash that survived the operators in early runs turned out, on analysis, to be an implementation error rather than a theory failure — the benchmark converged to the casebook's own verdict table.

6 Success quality and limits

Three qualifications keep the 100% honest.

First, the number measures the *implemented layers against their own targets*. The four NOT_COVERED cases and the one layer-flagged expectation are not noise; they delimit what a Baumgartner-layer engine can do, and they map the implementation roadmap: the AG setting/deviancy machinery would convert the mentor and extended-double-prevention cases and the remaining bogus-prevention expectation; RT-B would add the explanatory 'because' cases.

Second, the Normaliser ran on gold input. The reported capacity is that of the tabulation-analysis-judgement-learning chain; the capacity to produce Stage-I normalisations from free text is a separate measurement, with its own failure modes (factor individuation, field relegation, marking-factor discovery), and the architecture deliberately isolates it as an agent role so that it can be evaluated against the same casebook without touching the engine.

Third, two theoretical devices live in the adapter rather than the engine: directness-by-process-order and acyclicity-filtering. Both are sound on the benchmark but are stand-ins for structural-redundancy elimination (SM) and for a principled csf orientation. The wind-farm upstream biconditional is still *generated* by the engine (it is merely never misread), which is exactly the state of the pre-2023 regularity literature — the benchmark thus locates the engine, on the casebook's own map, between Baumgartner (2013a) and Baumgartner and Falk (2023).

7 The best-model criterion

What the system now treats as *the best model* deserves explicit statement, because it is the deterministic heart of everything reported above — and because it must not be conflated with probabilistic model selection. No clause below is a metric; there are no scores, no fit indices, no priors. The probabilistic layer of the framework (consistency/coverage thresholds, robustness, model averaging over ambiguity sets) is a *separate* stratum for noisy data and is not consulted here; where it is used at all, it operates over the survivors of this filter, never in place of it.

A *lexicographic filter* orders candidates the way a dictionary orders words: criteria are applied in strict priority, each criterion eliminates candidates outright, and no later criterion can overturn an earlier one. The contrast with a scoring function is deliberate and total — nothing is weighed, no criterion trades against another, no fit index is summed. A model that fails BM1 is gone before BM2 is consulted;

no degree of directness (BM4) compensates for a redundant conjunct (BM2). What survives every gate is best *simpliciter*; if several survive, that is reported as ambiguity, not broken by preference. This is why the criterion is not a probabilistic model-selection rule and must not be read as one: there is no likelihood, no penalty term, no averaging. The gates, in order:

BM1 — Admissibility (homogeneity). Best models are defined only over homogeneous mosaics, *relative to the outcome*: a configuration of the remaining factors observed with both values of E does not lower any score — it suspends interpretation and mandates expansion. Contradiction is a signal, not noise.

BM2 — Internal non-redundancy, mosaic-relative. Candidate theories are RDN-biconditionals whose minimal sufficiency is evaluated against the observed negative rows (not the full Boolean cube) and whose covers are irredundant — minimal necessity proper, not minimum cardinality. These are the two engine corrections of Section 3, now load-bearing.

BM3 — Orientability (CPLX). Only theories with at least two disjuncts orient the dependency by the asymmetry of determination. Single-route survivors without a process order are reported as dependencies with the causal reading withheld; with a supplied order they may be provisionally oriented, and the report says so explicitly.

BM4 — Directness. Among fit-equivalent candidates, admissible covers contain only factors strictly upstream of the outcome in the Stage-I process order, and the most proximate profile wins; downstream-mentioning biconditionals are excluded from the causal reading. This clause is marked *provisional*: it stands in for structural-redundancy elimination (Baumgartner and Falk 2023) until (SM) moves into the engine.

BM5 — Permanence. The best model is the model that survives expansion — the fixed point of the learning loop, not the winner of a single pass. Ties that survive BM1–BM4 are an *output*: honest underdetermination, resolvable only by further evidence (CU), never by preference. Best model ≠ unique model.

The criterion is implemented as `best_model(cases, outcome, order)` in `causal_architect.mt.best_model`, returning a report with one of five statuses — BEST_MODEL, UNDERDETERMINED, UNORIENTABLE, NO_THEORY, INHOMOGENEOUS — the selected theory, the surviving candidates, the excluded upstream biconditionals, and, where orientation rests on the order rather than on structure, an explicit flag. Unit tests pin the statuses to the casebook structures: the hooters mosaic yields the unique best model $T + B \Rightarrow L$; the chain is UNDERDETERMINED without an order and selects the direct theory $C + D \Rightarrow E$ with one; the reversed smoking–cancer direction is INHOMOGENEOUS (the strongest possible refusal: the putative outcome is not even a function of the remaining factors); the impoverished epiphenomenon view triggers BM1 and stabilises, after expansion, on $C + E \Rightarrow B$.

8 The paper as analytical target: LK-99

The casebook supplied golden-standard verdicts; the second deployment replaces them with a published experimental paper whose causal claims are the analytical target. The pipeline

(`benchmark/paper_lab.py`) is unchanged — normalise, tabulate, run the best-model criterion, judge, learn — but the Critic now adjudicates the *paper's own claims* against the evidence table assembled from the published record, and the learning loop plays the role the replication community actually played.

The target is Lee et al. (2023), the LK-99 room-temperature superconductivity claim — chosen because its resolution is, structurally, the casebook's paradigm case. Stage-I normalisation (agent-supplied, with provenance per row in the module): upstream, the composition factors *A* (Cu-doped lead-apatite phase) and *S* (Cu_2S impurity phase), and the state-typed factor *R* — the sample is in the superconducting state under the measurement conditions, i.e. has undergone a genuine superconducting transition; the background-physics alternative cause; downstream, the measurement outcomes *D* (sharp resistivity drop observed near 385 K) and *Z* (zero DC resistivity verified). The evidence rows are the published record: the original sample (Lee et al. 2023), pure Cu_2S with its first-order transition and three-to-four-orders resistivity drop at about 385 K (Zhu et al. 2023), phase-pure insulating single crystals free of Cu_2S (Puphal et al. 2023), the causal-field baseline, and a reference superconductor row supplying the alternative route to *D* and the only route to *Z*; the consensus reading of the episode follows Garisto (2023).

Stage	Outcome	Status	Best model	Notes
iter 1 (paper's view, S unmeasured)	D	INHOMOGENEOUS	—	2 contradictory configuration(s) for D: latent factor signalled; expansion mandated before any causal reading (BM1).
iter 1 (paper's view, S unmeasured)	Z	BEST_MODEL	R ⟹ Z	Single-route theory oriented by the Stage-I process order, not by the asymmetry of determination

Stage	Outcome	Status	Best model	Notes
				(BM3): provisional.
iter 2 (EXPAND(S): replications measured the Cu2S synthesis by-product)	D	BEST_MODEL	R + S ⟹ D	1 candidate(s) excluded as upstream- or downstream-mentioning relative to the process order (BM4).
iter 2 (EXPAND(S): replications measured the Cu2S synthesis by-product)	Z	BEST_MODEL	R ⟹ Z	1 candidate(s) excluded as upstream- or downstream-mentioning relative to the process order (BM4).; Single-route theory oriented by the Stage-I process order, not by the asymmetry of determination (BM3): provisional.

Table 2 Best-model run for lk99

Causal claim	Verdict	Ground
C1 (Lee et al.): the modified lead-apatite structure is causally responsible for the resistivity drop	NOT SUPPORTED	best model of D: R + S ⟹ D; A eliminated by minimal necessity once

Causal claim	Verdict	Ground
		S is measured — the apatite rides along with the impurity as the Manchester hooter rides along with five o'clock
C2 (Lee et al.): superconductivity is proved (zero-resistivity among the proof elements)	NOT SUPPORTED	best model of Z: R ⟹ Z; the sample row has Z = 0 and R = 0 in the verified record: no route from the sample to verified zero resistivity
C3 (replication resolution): the Cu2S impurity is causally relevant to the drop	SUPPORTED	S is a disjunct of the best model of D alongside the superconducting-state alternative R

The run reproduces the history of the episode in two iterations of the loop. On the paper's own factor set (*S* unmeasured), the corpus is INHOMOGENEOUS — Lee's sample and Puphal's crystal instantiate the same visible configuration with opposite outcomes. Irreproducibility, in best-model terms, *is* the BM1 signal of a latent factor. Expanding by the factor the replications measured yields the unique best model $R + S \Rightarrow D$: the resistivity drop has exactly two alternative causes — a genuine superconducting transition, or the Cu_2S impurity transition — and the Cu-doped apatite is eliminated by minimal necessity, riding along with the impurity precisely as the Manchester hooter rides along with five o'clock. The claim adjudication follows: the mechanism claim T1 is not supported; the superconductivity claim T2 fails at the token level (the best model of Z is $R \Rightarrow Z$, and the sample row has neither R nor Z); the community's resolution T3 is supported.

Three honesty notes. The normalisation and the row encodings are agent-supplied from the published record, with provenance stated per row — this run demonstrates the Normaliser role on a real paper, it does not yet automate it. The crisp-set idealisation is doing visible work: separating D from Z is what lets the analysis distinguish a "superconducting-like" drop from superconductivity, and that separation is itself an interpretive act. And the reference-superconductor row imports background physics into the causal field; without it the biconditional for D would be single-route and BM3 would withhold the orientation — the analysis is explicit about what it borrows.

9 Conclusion

The learning environment did what such an environment is for: its first runs falsified the implementation, not the theory; its operators, each licensed by a condition of the formal apparatus (PERM, marking, cAC), repaired exactly the cases the theory says they should repair; and its final score — complete replication of the layer-achievable casebook — is therefore evidence about the theory's implementability, not merely about one program. The residue is sharply circumscribed: one normative expectation and four cases wait on the deviancy and belief-revision layers, and one structural condition (SM) waits to move from the adapter into the engine. The casebook served as specification, regression suite, and curriculum at once — the role the synthesis paper projected for it.

References

Andreas, Holger, and Mario Günther. 2019. "On the Ramsey Test Analysis of 'Because'." *Erkenntnis* 84 (6): 1229–62. https://doi.org/10.1007/s10670-018-0006-8.

Andreas, Holger, and Mario Günther. 2024a. "A Lewisian Regularity Theory." *Philosophical Studies* 181: 2145–76. https://doi.org/10.1007/s11098-024-02149-z.

Andreas, Holger, and Mario Günther. 2024b. "A Regularity Theory of Causation." *Pacific Philosophical Quarterly* 105 (1): 2–32. https://doi.org/10.1111/papq.12447.

Baumgartner, Michael. 2013a. "A Regularity Theoretic Approach to Actual Causation." *Erkenntnis* 78 (S1): 85–109. https://doi.org/10.1007/s10670-013-9438-3.

Baumgartner, Michael. 2013b. "Detecting Causal Chains in Small-n Data." *Field Methods* 25 (1): 3–24. https://doi.org/10.1177/1525822X12462527.

Baumgartner, Michael, and Christoph Falk. 2023. "Boolean Difference-Making: A Modern Regularity Theory of Causation." *British Journal for the Philosophy of Science* 74 (1): 171–97. https://doi.org/10.1093/bjps/axz047.

Garisto, Dan. 2023. "LK-99 Isn't a Superconductor — How Science Sleuths Solved the Mystery." *Nature* 620: 705–6. https://doi.org/10.1038/d41586-023-02585-7.

Graßhoff, Gerd, and Michael May. 2001. "Causal Regularities." In *Current Issues in Causation*, edited by Wolfgang Spohn, Marion Ledwig, and Michael Esfeld. Mentis.

Lee, Sukbae, Ji-Hoon Kim, and Young-Wan Kwon. 2023. *The First Room-Temperature Ambient-Pressure Superconductor*. https://arxiv.org/abs/2307.12008.

Puphal, Pascal, M. Y. P. Akbar, Matthias Hepting, et al. 2023. *Single Crystal Synthesis, Structure, and Magnetism of* $Pb_{10-x}\ Cu_x(PO_4)_6\ O$. https://arxiv.org/abs/2308.06256.

Zhu, Shilin, Wei Wu, Zheng Li, and Jianlin Luo. 2023. *First Order Transition in* $Pb_{10-x}\ Cu_x(PO_4)_6\ O$ $(0.9 < x < 1.1)$ *Containing* $Cu_2\ S$. https://arxiv.org/abs/2308.04353.